PO-BACKED PROJECT CASE STUDY

A connected plant begins with the physical network.

Fibre backbone, Cat6 access, rack organisation and field installation scope at Alkem Laboratories, Ankleshwar.

MAY 2019

PO 4540033312 • MATERIALS

PO 4540033351 • FIELD WORK

D-TECH SOLUTION INTEGRATORS PRIVATE LIMITED



THE ENGAGEMENT

Two purchase orders, one physical layer

The 2019 records separate material supply from onsite cabling work at the Ankleshwar plant. This case study maps both streams into one conceptual solution.

01 MATERIALS

PO 4540033312 | 07 May 2019

Armoured fibre, Cat6 cable, racks, modules, pigtails, patch cords, conduit and user outlets.

02 FIELD SERVICES

PO 4540033351 | 08 May 2019

Fibre laying and splicing, conduit laying, Cat6 punching, rack dressing and one rack installation.

READING GUIDE

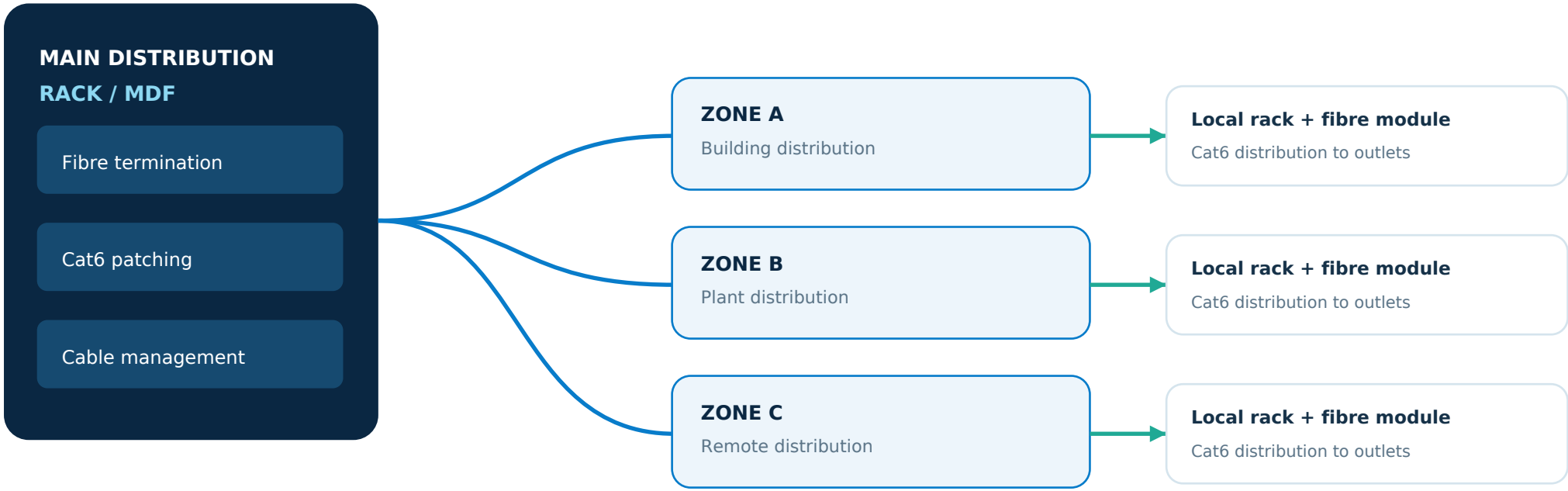
03 Site connectivity schematic
04 Rack and endpoint sketch
05 Delivery and verification path
06 PO scope register
07 Lessons and contact

SOURCE NOTE 2019 supplier on the POs: DTECH COMPUTERS. Current D-TECH branding is used for this portfolio study.

NETWORK CONCEPT

A traceable path from rack to user

The drawing translates the ordered physical components into a logical site architecture. It is illustrative, not an as-built plan.



ORDERED MATERIAL

770 m armoured fibre • 8 boxes Cat6 • 8 × 9U racks • 1 × 12U rack

The locations and switch/port assignments require the approved site drawings and as-built records.

Illustrative zones show connectivity relationships only; active switches are not specified in these two POs.

INSTALLATION DETAIL

From fibre route to a labelled outlet

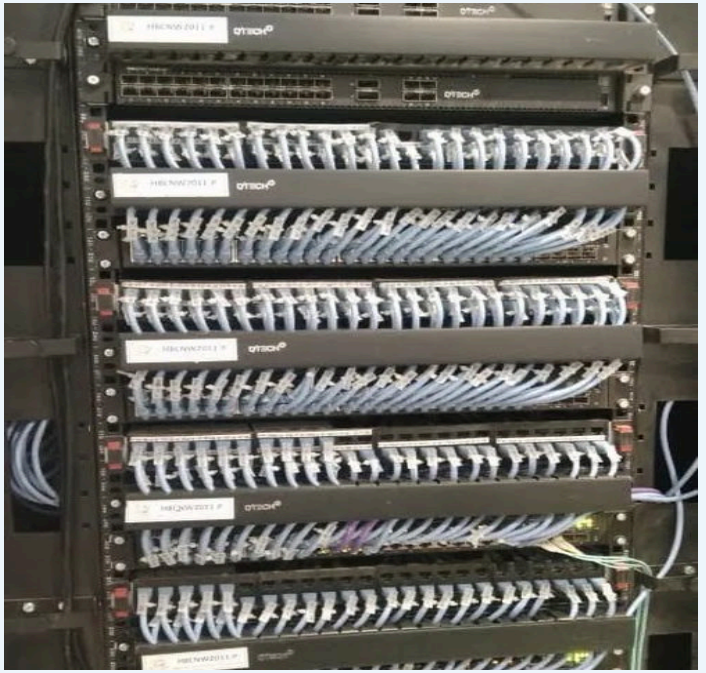
A clean installation links pathway, termination, patching and endpoint identification. The rack image was supplied with the source material.

01 ROUTE
Armoured fibre and 25 mm conduit, as scheduled

02 TERMINATE
Loaded fibre modules + pigtails + splicing

03 ORGANISE
9U/12U racks, cable managers and Cat6 panels

04 CONNECT
Patch cords to identified user-side Cat6 outlets



REFERENCE RACK PHOTO

Rack arrangement is a design illustration; the photograph alone does not establish a dated Alkem as-built record.

DELIVERY METHOD

Five stages from order to handover

The order lines identify the physical work. The checks below show how a site team can control quality and close the record.

01

Survey

Confirm routes, distances, rack positions and outlet locations.

02

Prepare

Stage fibre, Cat6, conduit, modules, racks and identifiers.

03

Install

Lay specified routes, mount racks and dress cables.

04

Terminate

Complete planned splices and Cat6 punching; label both ends.

05

Verify

Record test results, port maps and client acceptance.

FIELD WORK LISTED IN THE SERVICE PO

200 m	150 m	36	108	4
fibre laying	conduit laying	fibre splices	Cat6 I/O punches	racks dressed

Acceptance records, test reports and the final route schedule were not included among the supplied files.

ORDER REGISTER

The material and service quantities

Selected quantities are reproduced from the two Alkem purchase orders. Purchase quantities are not presented as installed quantities.

PO 4540033312 | MATERIALS

6-core armoured fibre	**770 m**
Cat6 cable	**8 boxes**
9U / 12U racks	**8 / 1**
Fibre modules (6 / 12 port)	**2 / 9**
Fibre pigtails	**120**
Cable managers	**11**
25 mm conduit	**500 m**
Cat6 I/O rack / user side	**180 / 30**
Cat6 patch cords 1 m	**126**
Fibre patch cords 3 m	**21**

PO 4540033351 | SERVICES

Fibre cable laying	**200 m**
25 mm conduit laying	**150 m**
Cat6 I/O punching	**108 nos**
Rack dressing / numbering	**4 racks**
Fibre splicing	**36 nos**
9U rack installation	**1 rack**

ENGINEERING CHECK Fibre designation and patch details in the PO remarks are mixed; confirm media and connector type on site.

WHAT THIS RECORD SHOWS

A practical physical network capability

The documents demonstrate the breadth of a plant connectivity package: backbone material, structured cabling, racks, terminations and onsite installation services.

01

ORDER SCOPE

Two complementary POs cover supply and field work at the same Ankleshwar site.

02

SOLUTION LOGIC

Fibre connects distribution points; Cat6, racks and patching connect local devices and users.

03

SERVICEABILITY

Numbered racks and documented terminations make tracing and future changes easier.

EVIDENCE BOUNDARY

The supplied records are purchase orders and reference images. The drawings are conceptual. Final installed quantities, test results and formal completion require separate site records.

Discuss a similar plant connectivity project

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